Christ College of Engineering, Irinjalakuda

Department of Electronics and Communicatiom

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GIDEON – Smart Offline Voice Assistant using Raspberry Pi

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Abstract

This project focuses on the design and implementation of an offline AI voice assistant using a Raspberry Pi. The main aim is to create a low-cost, portable intelligent assistant capable of understanding spoken questions and providing intelligent responses without relying on cloud services. The system integrates speech recognition, a locally hosted large language model (LLM), and text-to-speech synthesis to provide a complete human–computer interaction interface. The hardware components include a Raspberry Pi 5, USB webcam microphone for voice input, and Bluetooth speakers for audio output. The software stack consists of Python, Whisper for speech recognition, a TinyLlama LLM running locally through a llamafile server, and eSpeak-NG for voice output. The assistant records user speech, converts it to text, sends it to the LLM for response generation, and converts the response back into speech. The project demonstrates how embedded systems can be combined with artificial intelligence to create practical offline assistants. The outcome shows successful real-time interaction between the user and the AI assistant through voice commands. This system can be extended for applications such as smart home control, educational assistants, and accessibility tools**1.**

1. Introduction

Voice assistants such as Siri, Alexa, and Google Assistant require constant internet connectivity and cloud processing. This raise concerns related to privacy, latency, and dependency on network availability. There is a growing need for offline AI assistants that can operate locally on embedded devices.

With the advancement of edge AI and embedded computing, it is now possible to run lightweight language models locally. Raspberry Pi provides an affordable platform to experiment with AI integration in embedded systems. Such systems are useful in areas where internet connectivity is limited or privacy is important.

The main objectives of the project are:

- To design an offline AI assistant using Raspberry Pi
- To implement speech-to-text conversion
- To integrate a local LLM for intelligent responses
- To implement text-to-speech output
- To enable real-time human–machine interaction

2. Methodology / Design Approach

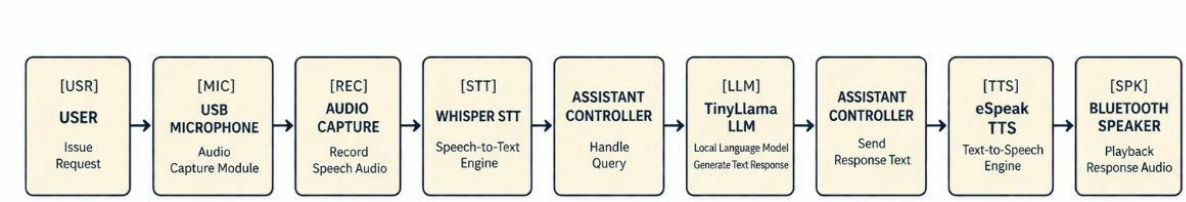
Materials and Tools Used

Hardware: Raspberry Pi 5, USB Webcam (Microphone input), Bluetooth Speaker, MicroSD card, Monitor and keyboard (for setup).

Software: Raspberry Pi OS (64-bit), Python, Whisper (Speech recognition), TinyLlama LLM, Llamafile server, eSpeak-NG (Text to speech), ALSA audio drivers**.**

The system was developed on a Raspberry Pi 5 running 64-bit Raspberry Pi OS. Voice input was captured via a USB microphone and transcribed using OpenAI's Whisper speech recognition model. The resulting text was processed by a locally hosted TinyLlama LLM served through a llamafile server, generating responses entirely offline. The response was then converted to speech using eSpeak-NG and delivered through a Bluetooth speaker.

Block Diagram



3. Results and Discussion

The system enables voice recognition, local AI responses, audio output, and continuous offline interaction with acceptable delay on Raspberry Pi. Streaming improves responsiveness. It is low-cost, portable, privacy-friendly, and expandable, though limited by CPU performance and audio setup complexity. Overall, it demonstrates that small language models can run effectively on embedded systems without cloud dependency.

4. Conclusion

The project demonstrates an offline AI assistant using Raspberry Pi by integrating speech recognition, local LLM processing, and speech synthesis for complete voice interaction, proving the feasibility of edge AI systems. However, it faces limitations like slower responses than cloud systems, hardware constraints, audio setup complexity, and limited memory. Future improvements include wake-word detection, enhanced voice synthesis, memory-based interaction, smart home integration, vision support, and GUI interfaces, with applications in assistants, education, accessibility, automation, and robotics.

References

[1] A. Radford et al., “Whisper: Robust Speech Recognition via Large-Scale Weak Supervision,” OpenAI Technical Report, 2022.

[2] H. Touvron et al., “LLaMA: Open and Efficient Foundation Language Models,” Meta AI Research, 2023.

[3] G. Gerganov, “llama.cpp: Inference of LLaMA Model in Pure C/C++,” GitHub Repository, 2023.